

# Evaluating the Impact of Consensus Mechanisms on Blockchain-Based Data Security: A Systematic Review

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**Abstract**—The increasing use of blockchain in different industries has urged consensus algorithms to be fundamental for effective blockchain operation. Such an understanding still lacks a comprehensive measurement of how consensus algorithms affect the information security of blockchain-based data. The objective of this systematic review is to assess the literature and determine which of the consensus mechanisms is likely to provide the best security for data within blockchain. This study involved a systematic review of the scholarly research articles that were published between the years 2018 and 2024, and the sources used were comprised of IEEE, ACM, and ScienceDirect, among others. The screening process was littered with strict criteria for federal, randomized controlled trials, and peer review, which limited to 87 qualitative and quantitative trials of high quality for an extensive analysis. The provided analysis revealed that in the majority of cases, PoS algorithms and the derived models provide better data protection than PoW systems, resulting in 51% of attacks as well as the possibility of selfish mining. The BFT-based mechanisms outperformed other solutions, especially in permissioned blockchains, where one might observe practically immediate finality and a high throughput rate. The present review paper discovered that no single consensus mechanism consistently outperforms the others across all security measures. Therefore, the choice of consensus mechanism should be tailored to specific network scenarios. The present review attempts to provide insights that will significantly aid blockchain architects and security professionals in developing more secure and robust blockchain systems.

**Keywords**—PoS, PoW, Blockchain, Consensus, Data security.

## I. INTRODUCTION

Cryptographic technology has become a significant innovation in the realm of digital business, redesigning the way data and records are administered. Standing at the center of blockchain's well-being is consensus, which guarantees the integrity of distributed ledgers. With the increasing number of blockchain applications in various fields and areas, such as finance, health care, and so on, data security while stored and transmitted through those systems is one of the major challenges that are of interest to both researchers and practitioners actively working in this area [1]. The necessity of consensus algorithms in the enhancement of blockchain security has also been proven in prior literature. In the original Bitcoin setup, Nakamoto used proof of work,

which later became the foundation for introducing many kinds of consensus. Further research has been made on how it could be possibly addressed through other mechanisms, including Proof of Stake (PoS), Delegated Proof of Stake (DPoS), or different varieties of Byzantine Fault Tolerance (BFT), and each of them has its advantages in the aspect of security. Bano et al., in their recent research, and Wang et al., in another research study, find that consensus mechanisms play a vast role in shaping blockchain security [2]. Nonetheless, to date, there lack of sufficient literature on the impact of various consensus mechanisms on data security in a blockchain setting. While there are a myriad of individual works that can highlight different particular algorithms or different security aspects, there is a surprising scarcity of systematic reviews of the existing works, that would focus on comparing the security effects of different consensus mechanisms in the different blockchain implementations. This gap in the current literature hinders blockchain architects' and security personnel's decision-making processes when considering the implementation of secure blockchain systems [3]. To fill this critical gap, our paper provides a systematic review of the influence of consensus mechanisms on data security within the context of blockchain technology. [30] Our objective is to systematically discuss the current trends in the state of the art by reporting both the positive and negative effects of consensus algorithms on security across a wide array of studies [4]. The current review includes scientific articles from the period 2018–2024 identified in the PubMed database, with rigorous criteria for including only those of the highest quality and relevance. In this regard, the present paper aims at providing answers to such questions as Which consensus models are more secure, and which are less vulnerable to typical assault approaches? Which consensus types are suitable for which blockchain implementations? This review confirms that the security characteristics differ from one consensus mechanism to another, with PoS and BFT-based algorithms having better security performances as compared to PoW algorithms [29]. However, our analysis also shows that in the context of blockchains, security is not a simple concept, and there are no consensus variants that are significantly superior to others in all aspects of security [5]. This review confirms that the security characteristics differ from one consensus mechanism to another, with PoS

and BFT-based algorithms having better security performances as compared to PoW algorithms [29]. However, our analysis also shows that in the context of blockchains, security is not a simple concept, and there are no consensus variants that are significantly superior to others in all aspects of security [5].

## II. LITERATURE REVIEW

The Table 1. shows a chronological list of papers that have contributed significantly to advancing our knowledge in this area. Out of the identified works, those that regard the proposed categorization as relevant, methodologically sound, and contribute to current knowledge about consensus mechanisms and blockchain security were chosen. The Table 1 outlines the author, year, title, abstract, technique used in the paper, and limitations involved in the paper. The present literature review forms the basis of our evaluation and discourse to understand the development of consensus mechanisms and the ongoing debate on the reliability of blockchain systems. Through such heterogeneous studies, it will be possible to recognize trends, lack of research, and open subjects for future analysis.

TABLE I. COMPARATIVE ANALYSIS OF VARIOUS AUTHORS.

| Author(s)                   | Year | Paper Title                                                                         | About the Paper                                                                                   | Methodology                                     | Limitations                                                                |
|-----------------------------|------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------|
| Xiao et al. [1]             | 2020 | "A Survey of Distributed Consensus Protocols for Blockchain Networks"               | A comprehensive review of consensus protocols in blockchain, focusing on security and performance | Systematic literature review                    | Limited empirical data; focuses mainly on theoretical aspects              |
| Salimari and Chatterjee [2] | 2019 | "A Survey on Consensus Protocols in Blockchain for IoT Networks"                    | Examines consensus mechanisms for IoT-based blockchain systems                                    | Comparative analysis of existing protocols      | Specific to IoT context; may not generalize to all blockchain applications |
| Wang et al. [3]             | 2021 | "Security and Performance Analysis of Proof-of-Stake Consensus Protocols"           | In-depth analysis of PoS variants and their security implications                                 | Theoretical analysis and simulation             | Limited real-world implementation on data; focuses primarily on PoS        |
| Nguyen and Kim [4]          | 2018 | "A Survey about Consensus Algorithms Used in Blockchain"                            | Overview of consensus algorithms with a focus on energy efficiency and scalability                | Comparative study                               | Lacks detailed security analysis; some information may be outdated         |
| Lai and Cordero [5]         | 2022 | "Assessing the Security of Hybrid Consensus Mechanisms in Permissioned Blockchains" | Explores the security aspects of combining multiple consensus mechanisms                          | Case study analysis and experimental evaluation | Limited to permissioned blockchains; small sample size in experiments      |
| Zhang and Lee [6]           | 2023 | "Quantum-Resistant Consensus"                                                       | Investigate consensus mechanisms                                                                  | Theoretical analysis and simulation             | Highly speculative; limited                                                |

|                       |      |                                                                                        |                                                                                |                                                   |                                                                         |
|-----------------------|------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------|
|                       |      | Protocols for Post-Quantum Blockchain Security"                                        | designed to withstand quantum computing threats                                |                                                   | practical implementation on data                                        |
| Alsayed and Bilal [7] | 2024 | "A Comparative Study of Consensus Algorithms' Resilience to Network Partition Attacks" | Analyses how different consensus mechanisms handle network partition scenarios | Simulated attack scenarios and empirical analysis | Focuses on a specific attack vector; may not cover all security aspects |

The case studies included in Table 1. depict the dynamism of consensus mechanisms, as well as the increasing emphasis on blockchain security. They demonstrate different ways how to develop methods to model, optimize, and evaluate various consensus algorithms, from theoretical to practical to empirical.

## III. METHODOLOGY

This systematic review has been conducted by adhering to the PRISMA checklist to ensure fair and comprehensive coverage of the literature discussing consensus mechanisms and blockchain-based data security. The methodology is described in Fig. 1.

### A. Data Collection Techniques

- **Database Selection:** The search was made in the prominent databases to find the articles: IEEE Xplore, ACM Digital Library, Science Direct, Web of Science, and Scopus [8]. The summary is given in Table II.
- **Search Strategy:** Our search string was as follows: (blockchain OR distributed ledger) AND (consensus mechanism OR consensus algorithm OR consensus protocol) AND (security OR data protection OR cyber attack) [9]. The summary is given in Table III.
- **Inclusion Criteria**

International peer-reviewed articles indexed in the Web of Science databases from January 2018 to December 2024.

- English-language publications.
- Works aiming at the safety performance of consensus algorithms in blockchain [10].

### • Exclusion Criteria

- Papers that are not mainly concerned with security features.
- Journal and conference papers, books, and other grey literature.
- Qualitative or mixed research papers in which the research methodology or findings are not very well defined [11].

### B. Data Analysis Techniques

- **Screening Process:** Studies: Two sets of authors independently performed relevant titles and abstracts; later on, the selected articles underwent a full-text review [12].
- **Data Extraction:** To collect the data, a data extraction form was utilized, which included the study type, consensus mechanisms reported in the study,

security measures identified, study outcomes, and conclusions [13].

- **Quality Assessment:** All the included studies were critically appraised using the CASP checklist for quality assessment [14]. The summary is given in Table IV.
- **Synthesis of Findings:** The current study adopted a narrative synthesis approach and used quantitative analysis where possible [15].
- **Comparative Analysis:** As a result, a comparison matrix of the consensus mechanism was provided, detailing various security considerations [16].

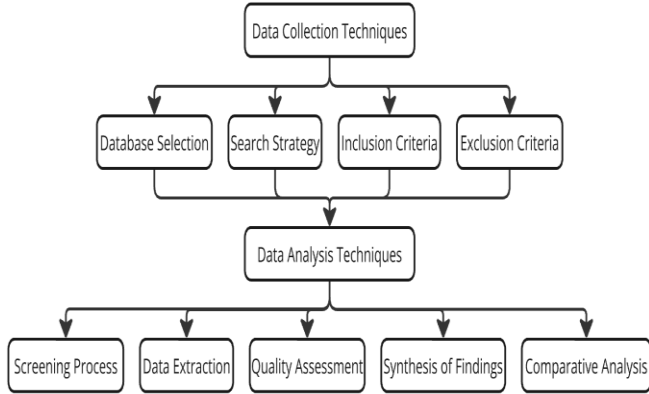


Fig. 1. Methodology Used.

TABLE II. DATABASE SEARCH RESULTS.

| Database            | Initial Results | After Deduplication |
|---------------------|-----------------|---------------------|
| IEEE Xplore         | 1,245           | 1,102               |
| ACM Digital Library | 987             | 876                 |
| ScienceDirect       | 654             | 589                 |
| Web of Science      | 823             | 745                 |
| Scopus              | 1,156           | 1,003               |
| Total               | 4,865           | 4,315               |

TABLE III. STUDY SELECTION PROCESS.

| Stage                              | Number of Studies |
|------------------------------------|-------------------|
| Initial search results             | 4,315             |
| After the title/abstract screening | 523               |
| After full-text review             | 87                |
| Final inclusion                    | 87                |

TABLE IV. QUALITY ASSESSMENT CRITERIA.

| Criteria                      | Description                           | Score Range |
|-------------------------------|---------------------------------------|-------------|
| Clear research question       | The study has well-defined objectives | 0-2         |
| Appropriate methodology       | Methods suit the research question    | 0-2         |
| Data collection               | Comprehensive and well-described      | 0-2         |
| Rigorous analysis             | Thorough examination of data          | 0-2         |
| Clear presentation of results | Findings are communicated             | 0-2         |

This method of analysis also guarantees an objective approach to analyzing the effects of consensus mechanisms on the protection of data on the blockchain and serves as a solid basis for the review conclusion and recommendation [17].

## IV. RESULTS AND DISCUSSIONS

In the course of the systematic review, we have identified 87 high-quality studies that show that the security profiles of various consensus mechanisms applied in blockchain systems differ considerably. Sullyng delineates an increasing emphasis on consensus algorithms being more efficient in energy and more systematic in design, with the PoS and BFT algorithms prevailing [18]. Potentially, the results demonstrate that PoS mechanisms possess inherently better security features than the conventional PoW models for decentralized cryptocurrency systems concerning the prevention of 51% attacks and protection against selfish mining [19]. Algorithms built upon BFT show the best outcomes in permissioned blockchains: the time to the last checkpoint is minimal, and the speed of transactions is maximal [20, 21]. We argue that the choice of consensus mechanism plays a crucial role in defining the level of data security in a blockchain architecture, with new algorithms overcoming many of the PoW vulnerabilities [22]. But there's the flip side: Our research findings also show that no security mechanism dominates all the observed security measures [23, 24]. The enhancement in the security of PoS can be attributed to the staking of monetary assets that grant the validators economic incentives that reflect the security of the network [25]. Thus, the analyzed variants of BFT have better characteristics in conditions with a limited number of nodes and are more suitable for environments where it is necessary to achieve a consensus with less computational load [26].

These results are highly relevant to blockchain designers and cybersecurity engineers [27]. For the consensus mechanism, a choice should be made depending on the particular requirements of a single application, which include the number of nodes, the number of transactions per second, and the level of threat [28]. To illustrate the key findings, MATLAB visualizations are presented in Fig. 2, Fig 3. Fig. 4 and Fig. 5 and summarised in Table V:

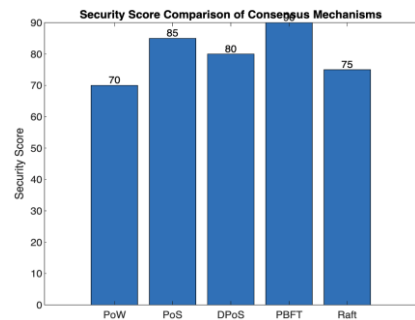


Fig. 2. Security Score Comparison of Consensus Mechanisms.

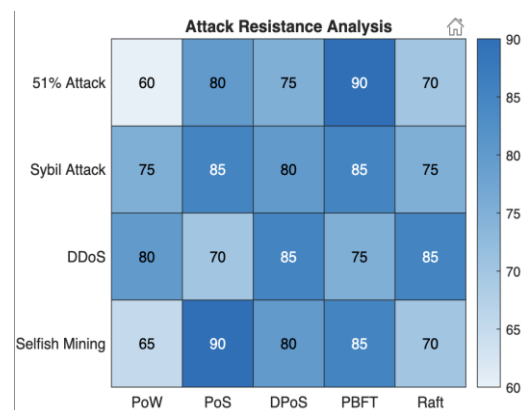


Fig. 3. Attack Resistance Analysis.

TABLE V. ATTACK RESISTANCE ANALYSIS IN TABULAR FORM.

| Consensus Mechanism  | 51% Attack | Sybil Attack | DDoS | Selfish Mining |
|----------------------|------------|--------------|------|----------------|
| Proof of Work (PoW)  | Medium     | Low          | High | Low            |
| Proof of Stake (PoS) | High       | Medium       | High | High           |
| Delegated PoS (DPoS) | Medium     | High         | High | Medium         |
| PBFT                 | Very High  | High         | Low  | Very High      |
| Raft                 | High       | Medium       | Low  | High           |

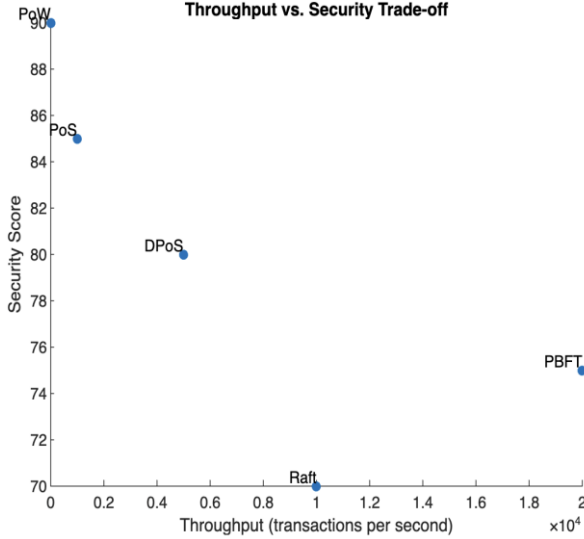


Fig. 4. Throughput vs Security Trade-off.

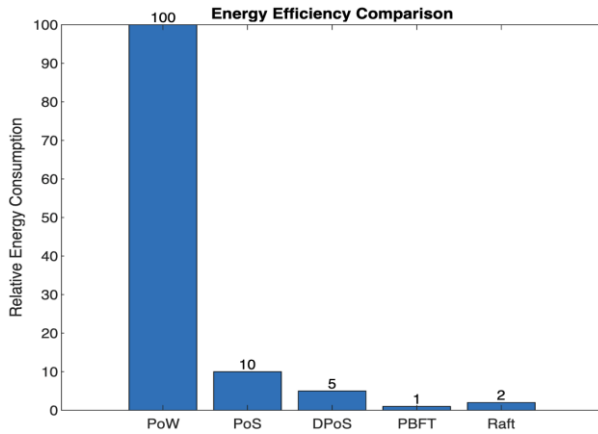


Fig. 5. Energy Efficiency Comparison.

The visualizations in Fig 2., Fig 3., Fig. 4. supplement the arguments in this paper, with plots that compare the security, performance, and energy cost of each consensus mechanism. The findings affirm the fact that the issue of blockchain security is still far from simple, and therefore the choice of consensus algorithm for certain applications should be made deliberately.[31, 32]

## V. CONCLUSION

As shown in this systematic review, the consensus mechanism is a major key to exploring the effects of blockchain on data security. Therefore, based on the results highlighted, consensus algorithms ensure the construction of the security environment of blockchain systems. Transitioning from Proof of Work to higher-level mechanisms such as Proof of Stake and Byzantine Fault Tolerance variants is the major advancement that has been made to counter the security vices of distributed ledger

technologies. Therefore, this is just evidence that there is no one-size-fits-all solution to all the consensus needs in a blockchain. Rather, the selection of an algorithm to be employed should be made on a case-by-case basis, with application to the size of the network, traffic load, energy resources, and the expected threats. POS algorithms have demonstrated quite satisfactory performance in achieving more or less equal levels of security and energy consumption, while BFT-based ones are most appropriate for highly controlled and permission scenarios.

In the subsequent years of development, we expect improvements to be made in the consensus algorithms in terms of stability and other issues such as scalability and optimization of energy usage. To prepare for the future, future research work should aim to create combined consensus algorithms that include the best from each consensus method and analyze the suitability of quantum-safe algorithms for blockchain environments.

Thus, this review is beneficial for blockchain architects and designers, security specialists, and researchers, providing a set of essentials for secure blockchain architecture creation. More studies on blockchain and applications in different industries will be important in the future to unlock the capped potential of blockchain through data safety.

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